



# Drop

Clean Water, Delivered to Your Door

On-demand delivery platform for neighborhood water refill stations.  
Scan nearby stations, compare refill prices, and eliminate the 20kg hauling struggle.

# The Problem: Water Refill Station Logistics

Millions of urban Kenyans rely on local refill stations for clean 20L drinking water. But getting it home is exhausting: walking with empties, carrying 20kg back, or bargaining with random street bodas.



## The Exhausting "20kg Walk"

To get clean drinking water, customers must carry an empty 20L bottle, walk to a neighborhood refill shop, wait in line, and physically haul a back-breaking 20kg (44 lbs) refilled container back home on foot — often climbing several flights of apartment stairs.



## The Inconvenient Boda Compromise

To avoid the 20kg walk, customers must walk to a roadside stage, hail a motorbike rider, haggle over arbitrary fares, and dangerously balance a 20L bottle on the bike — turning an affordable KSh 80 refill into an expensive, risky ordeal.



## Refill Stations Lack Delivery Fleets

Neighborhood water refill shops and Automated Teller Machines (ATMs) have pure purification capacity, but operate strictly as walk-in counters. With no delivery fleet or dispatch tools, they cannot reach busy families, offices, or customers living beyond walking distance.



## Blind Discovery & Price Opacity

Customers cannot easily scan or compare nearby refill stations (prices vary from KSh 60 to KSh 120+). They have no way to verify station operating status, track queue times, or coordinate reliable empty bottle exchanges.

# The Solution: On-Demand Refill Delivery

Drop bridges the gap between water refill stations and thirsty households — scan nearby stations, compare prices in real time, and get heavy 20L bottles delivered straight to your door.



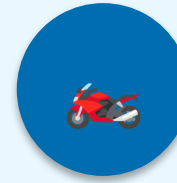
## Customer App

Scan nearby refill stations on the map, compare live prices, order 20L refills with one tap, pay via M-Pesa, and get heavy bottles delivered straight to your door.



## Vendor App

Equips stationary water refill shops with a turnkey on-demand delivery arm — manage orders, track bottle inventory, dispatch riders, and grow far beyond walk-in traffic.



## Rider App

Enables motorbike riders to accept delivery jobs, collect customer empties, exchange them at partner refill stations, and earn steady income handling the heavy hauling.

+ Admin Operations Console — 27 screens for refill station management, bottle float tracking, live dispatch, and automated financial settlements.

# Market Opportunity

**TAM**

**Total Addressable Market**

Scope: Entire Kenya Water Economy

**\$3.2B**

Total annual expenditure on water supply & sanitation across Kenya — formal public utilities, private tankers, water kiosks, and vendors (WASREB & World Bank estimates).

**SAM**

**Serviceable Available Market**

Scope: Urban 20L Refill & Packaged Water

**\$850M**

Annual market for packaged and vended 20L refill water across Kenya's urban and peri-urban centers, serving households and businesses requiring clean drinking water.

**SOM**

**Serviceable Obtainable Market**

Scope: Nairobi Metro Beachhead (24 Mo)

**\$45M**

Immediate 24-month serviceable capture: 1.8M households across Nairobi metro and high-density satellite towns (Rongai, Ngong, Ruiru, Kiambu) ordering 2–4 refills/week.

**54M+**

**Kenya Population**

KNBS & World Bank (2024/25)

**83.5%**

**Smartphone Adoption**

Communications Authority (CA)

**34M+**

**Active M-Pesa Customers**

Safaricom Annual Report FY24/25

**58%**

**Rely on Non-Tap / Refills**

WASREB 18th Report (42% piped safe)

 All metrics verified against official sources: Water Services Regulatory Board (WASREB) 18th Impact Report • Safaricom PLC Annual Reports • Communications Authority of Kenya (CA) • KNBS

# How It Works: Doorstep "Swap & Drop" Flow

A frictionless 6-step physical exchange loop that eliminates hauling heavy 20kg water bottles up stairs.

1

## Scan & Discover

Customer opens Drop, scans live neighborhood water refill stations on the map, compares prices (KSh 60–100), and reviews station ratings.

2

## Order & M-Pesa Pay

Customer selects 20L refill quantity, specifies empty bottle return count, and checks out via seamless M-Pesa SIM Tool Kit (STK) Push.

3

## Smart Dispatch

Two-tier dispatch engine broadcasts the delivery to station-employed riders first, or to verified neighborhood gig riders within seconds.

4

## Exchange & Haul

Rider collects customer empty 20L bottles from the doorstep, swaps for purified bottles at the partner station, and handles the heavy lifting.

5

## Doorstep Delivery

Customer tracks rider Global Positioning System (GPS) in real time. Rider climbs stairs and delivers 20kg bottles directly into the kitchen.

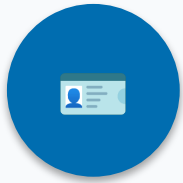
6

## Instant Settlement

M-Pesa escrow splits revenue instantly: vendor and rider wallets receive payouts, while the digital bottle ledger updates with zero deposit dispute.

# Platform Defensibility & Anti-Leakage

Why customers, riders, and refill stations stay locked onto Drop instead of transacting off-platform.



## Digital Bottle Float Ledger

Each 20L container costs KSh 450–600. Vendors lose significant capital annually to stolen or unreturned bottles. Drop's cryptographically audited custody ledger and escrow deposit guarantee eliminate bottle loss disputes — vendors refuse off-platform deliveries to protect their float.



## Elastic Multi-Vendor Fallback

Neighborhood refill stations frequently suffer borehole rationing, power blackouts, or filter maintenance. Direct offline customer-vendor relationships fail when a shop goes dry. Drop dynamically routes orders to the nearest open station, guaranteeing delivery in under 45 minutes.



## Automated M-Pesa Escrow

Direct offline arrangements require awkward cash change or manual M-Pesa transfer haggling on the stairs. Drop eliminates payment friction: customers pay via one-click prompt, while riders and vendors receive automated, dispute-free wallet payouts upon verified delivery.



## Rider Batching & High Earnings

Independent riders make 40% more income on Drop by batching multiple heavy 20kg deliveries along clustered estate routes. During slow mid-day retail hours, Drop provides steady on-demand delivery volume that no single physical shop can offer alone.

# Business Model

Multiple revenue streams, all configurable in real time from the admin console — 34 pricing parameters across 10 groups.

## Vendor Commission

**5% on retail  
(Business-to-Consumer)  
5% on wholesale  
(Business-to-Business)**

Scales with every order on the platform

## Service Fees

**KSH 35 per retail order  
KSH 120 per wholesale order**

Covers M-Pesa tariff + platform ops costs

## Surge Pricing

**+KSH 10 during peak hours (6-8am, 5-7pm)**

Dynamic pricing for high-demand windows

## Delivery Fees

**Distance-based rider earns 90-100%**

5% wholesale markup; retail markup at 0%

 One pricing path: `compute_order_quote` is the single source of truth — client renders it, M-Pesa charges it, database stores it. Automated tests enforce parity.

# Predictable, Profitable Unit Economics

Drop achieves healthy positive unit contribution margin on every single 20L bottle delivered from Day 1.

| Retail Order: 1x 20L Refill Bottle              |                          | Wholesale Order: 20x 20L Bottles              |                           |
|-------------------------------------------------|--------------------------|-----------------------------------------------|---------------------------|
| Typical Household Order (2 km average distance) |                          | Office / School Bulk Order (Tuk-Tuk, 5 km)    |                           |
| Water Refill Price (Station)                    | KSh 100.00               | Wholesale Water (20 × KSh 70)                 | KSh 1,400.00              |
| Delivery Fee (Base 80 + 20/km × 2 km)           | KSh 120.00               | Freight Delivery (Tuk-Tuk: 150 + 100×5km)     | KSh 650.00                |
| Platform Service Fee                            | KSh 35.00                | Platform Delivery Markup (5% of Fee)          | KSh 32.50                 |
| Total Paid by Customer (M-Pesa)                 | KSh 255.00               | Wholesale Platform Service Fee                | KSh 120.00                |
| Vendor Net (Refill Price less 5%)               | KSh 95.00                | Total Paid by Commercial Client               | KSh 2,170.00              |
| Rider Net (Delivery less 10% Commission)        | KSh 108.00               | Vendor Net (Product – 5% + Delivery – Markup) | KSh 1,947.50              |
| <b>Gross Platform Revenue (5 + 12 + 35)</b>     | <b>KSh 52.00</b>         | <b>Gross Platform Take (70 + 32.50 + 120)</b> | <b>KSh 222.50</b>         |
| M-Pesa Collection Cost (STK Push)               | -KSh 15.00               | M-Pesa Collection Cost                        | -KSh 15.00                |
| <b>Net Platform Contribution Margin</b>         | <b>KSh 37.00 (71.2%)</b> | <b>Net Platform Contribution Margin</b>       | <b>KSh 207.50 (93.3%)</b> |

 Compounding Flywheel: 4.2 monthly refills per household = KSh 155 monthly net contribution per active customer. Customer Acquisition Cost (CAC) estimated at KSh 180 is fully paid back in under 35 days.



# Technology Stack

## BACKEND

- ▶ Python 3.12 • FastAPI Framework • Uvicorn
- ▶ PostgreSQL Database + PostGIS + H3 Spatial Indexing
- ▶ SQLAlchemy 2.0 Async Object-Relational Mapping • Alembic
- ▶ Redis (Publish/Subscribe, Job Queues, Rate Limiting)
- ▶ ARQ Asynchronous Background Job Workers
- ▶ M-Pesa STK (SIM Tool Kit) Push / Business-to-Consumer (B2C) / Reversals
- ▶ Amazon Web Services (AWS) Simple Storage Service (S3)
- ▶ Clerk RS256 JSON Web Token (JWT) Authentication
- ▶ Sentry Error Monitoring • Prometheus Metrics • JSON Logs

## MOBILE & WEB

- ▶ Expo Software Development Kit (SDK) 54 • React Native 0.81
- ▶ React 19 • Expo Router 6 (File-Based Navigation)
- ▶ NativeWind v4 Styling • TanStack Query v5 Data Fetching
- ▶ Zustand v5 State Management Architecture
- ▶ @clerk/clerk-expo Client Authentication
- ▶ react-native-maps + Live WebSocket Location Streaming
- ▶ Next.js 16 (Operations Console Web Application)
- ▶ Tailwind CSS v4 • Recharts Visualizations
- ▶ Vercel Cloud Deployment (Operations Console + Website)



Architecture Principle: One backend, one database, one pricing path. The console is a Backend-for-Frontend (BFF) — the browser never holds an Application Programming Interface (API) token.

Geospatial: Uber H3 spatial hex indexing at resolution 8 with PostGIS ST\_DWithin for exact-distance discovery.

# Operations Console — 27 Screens

Every page carries aggregates that tell an operator whether anything needs doing — not just tables with search boxes.

## Operations

Orders board • Know Your Customer (KYC) Verification • Live Map Catalogue • Bottle Float Ledger • Vendor Verification • Disputes Reviews • Delivery Replay

## People

Performance Dashboard • Fleet Management Customers • Riders • Vendors

## Finance

Payouts • Transactions Reconciliation • Settlement

## Support

Ticket Inbox  
Threaded Conversations

## Platform

Administrators • Audit Log • Notifications  
Broadcast • Pricing (34 settings) • Settings

# Security & Guardrails

Platform-wide invariants enforced by tests that fail the build — not just documented rules.



## Financial Integrity

Money is Decimal, never float — from database through Application Programming Interface (API) to formatted string. One quote, one charge, one stored total.



## Auth & Access Control

Clerk RS256 JSON Web Token (JWT) verification. Every order endpoint calls `authorise_order_access`. 26 admin capabilities, role-based presets.



## Proof of Delivery

Photo proof mandatory on bottle deficit. Verification wall blocks riders until Know Your Customer (KYC) approved. Identity documents presigned for 5 min.



## Zero-Trust Console

Backend-for-Frontend (BFF) architecture — browser never holds Application Programming Interface (API) token. No retry on reversals. Every mutation audited with written reason.



## Concurrency Safety

FOR UPDATE ... SKIP LOCKED on all sweeps. Commit-per-item. Two sanctioned push paths only — bare `create_task` fails the build.



## Structural Tests

704 tests including Abstract Syntax Tree (AST) walkers that reject: unguarded routes, missing capability checks, bypassed proof checks, retry buttons on settlements.

# Competitive Advantage

| Feature                                            | Drop     | WhatsApp Vendors | Maji.com | Traditional |
|----------------------------------------------------|----------|------------------|----------|-------------|
| Real-time GPS (Global Positioning System) tracking | ✓        | ✗                | ✗        | ✗           |
| M-Pesa integration                                 | ✓        | Manual           | ✓        | Cash only   |
| Multi-vendor marketplace                           | ✓        | ✗                | Single   | ✗           |
| Proof of delivery                                  | ✓ Photo  | ✗                | ✗        | ✗           |
| Bottle deposit tracking                            | ✓ Ledger | ✗                | ✗        | ✗           |
| Admin console (27 screens)                         | ✓        | ✗                | Basic    | ✗           |
| Smart two-tier dispatch                            | ✓        | ✗                | ✗        | ✗           |
| Revenue reconciliation                             | ✓ Auto   | ✗                | Manual   | ✗           |

# Go-to-Market Strategy

## Phase 1: Seed Beachhead

Months 1–6

- ▶ Focus corridor: Ngong Town & Rongai
- ▶ Onboard 50 partner refill stations
- ▶ Recruit 100 vetted gig riders
- ▶ 1,000 active household customers
- ▶ Validate unit contribution margin

## Phase 2: Metro Expansion

Months 7–18

- ▶ Expand to 5 Nairobi sub-counties
- ▶ Scale to 500 vendors, 1,000 riders
- ▶ Launch wholesale (Business-to-Business)
- ▶ Office & school recurring subscriptions
- ▶ Target 25,000 monthly delivered orders

## Phase 3: National Scale

Months 19–36

- ▶ 10 major Kenyan urban hubs (Mombasa, Kisumu)
- ▶ White-label corporate dispatch portal
- ▶ Water quality certification testing badges
- ▶ Regional East Africa expansion (Uganda & Tanzania)
- ▶ Target 200,000+ monthly orders

# Financial Projections & Mathematical Derivation

Every metric is mathematically derived from Slide 8 platform unit economics (Retail AOV: KSH 255 | Wholesale AOV: KSH 2,170 | M-Pesa: KSH 15/tx).

| Year 1<br>30k Annual Orders (90% Retail / 10% Wholesale)                                                                 | Year 2<br>180k Annual Orders (85% Retail / 15% Wholesale)                                                                  | Year 3<br>1.2M Annual Orders (80% Retail / 20% Wholesale)                                                                    |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Monthly Orders (Run-rate)<br><b>5,000</b><br><small>Formula: Ramp avg 2,500/mo × 12 = 30,000 orders/yr</small>           | Monthly Orders (Run-rate)<br><b>25,000</b><br><small>Formula: Ramp avg 15,000/mo × 12 = 180,000 orders/yr</small>          | Monthly Orders (Run-rate)<br><b>200,000</b><br><small>Formula: Ramp avg 100,000/mo × 12 = 1,200,000 orders/yr</small>        |
| Gross Merchandise Value (GMV)<br><b>KSH 13.4M</b><br><small>Formula: (27k retail × 255) + (3k wholesale × 2,170)</small> | Gross Merchandise Value (GMV)<br><b>KSH 97.6M</b><br><small>Formula: (153k retail × 255) + (27k wholesale × 2,170)</small> | Gross Merchandise Value (GMV)<br><b>KSH 765.6M</b><br><small>Formula: (960k retail × 255) + (240k wholesale × 2,170)</small> |
| Platform Take<br><b>KSH 2.07M</b><br><small>Formula: (27k retail × 52) + (3k wholesale × 222.50)</small>                 | Platform Take<br><b>KSH 14.0M</b><br><small>Formula: (153k retail × 52) + (27k wholesale × 222.50)</small>                 | Platform Take<br><b>KSH 103.3M</b><br><small>Formula: (960k retail × 52) + (240k wholesale × 222.50)</small>                 |
| Payment Costs<br><b>KSH 0.45M</b><br><small>Formula: 30,000 total orders × KSH 15 M-Pesa fee</small>                     | Payment Costs<br><b>KSH 2.7M</b><br><small>Formula: 180,000 total orders × KSH 15 M-Pesa fee</small>                       | Payment Costs<br><b>KSH 18M</b><br><small>Formula: 1,200,000 total orders × KSH 15 M-Pesa fee</small>                        |
| Operating Costs (OPEX)<br><b>KSH 3.5M</b><br><small>Formula: Core dev + cloud servers + seed operations</small>          | Operating Costs (OPEX)<br><b>KSH 12M</b><br><small>Formula: Nairobi-wide expansion, support &amp; fleet ops</small>        | Operating Costs (OPEX)<br><b>KSH 65M</b><br><small>Formula: Multi-city scale (Mombasa, Kisumu, Nakuru)</small>               |
| Net Contribution<br><b>-KSH 1.88M</b><br><small>Formula: Take (2.07M) - Payments (0.45M) - OPEX (3.5M)</small>           | Net Contribution<br><b>-KSH 0.7M</b><br><small>Formula: Take (14.0M) - Payments (2.7M) - OPEX (12.0M)</small>              | Net Contribution<br><b>KSH 20.3M</b><br><small>Formula: Take (103.3M) - Payments (18.0M) - OPEX (65.0M)</small>              |

MASTER FORMULAS: GMV = Σ(Orders × AOV) | Take = (Retail Orders × KSH 52) + (Wholesale Orders × KSH 222.50) | Payments = Total Orders × KSH 15 | Net = Take - Payments - OPEX

# Current Traction

## **Product Built**

Full-stack platform complete — 3 mobile apps, admin console, backend API, all deployed and live.

## **Beta Live**

Android Application Packages (APKs) available for download on website. Beta credentials active for external testers.

## **704 Tests Passing**

Comprehensive test suite with Abstract Syntax Tree (AST) structural walkers. Continuous Integration (CI/CD) on every push.

## **27 Admin Screens**

Operations console with Know Your Customer (KYC), disputes, reconciliation, delivery replay, and financial controls.

## **Security Hardened**

Clerk authentication, Backend-for-Frontend (BFF) pattern, encrypted Personally Identifiable Information (PII), audited mutations.

## **Fully Documented**

12 architecture docs, decision records, runbooks, and deployment guides.

# Leadership & Operational DNA

A purpose-built team pairing institutional software engineering with grassroots Kenyan trade logistics.



## Technical & Systems Architecture

### Full-Stack Lead & Platform Architect

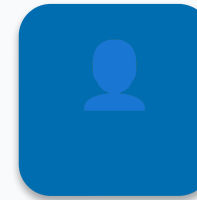
Expert in high-reliability distributed systems, geospatial H3 indexing, and financial state machines. Designed and built the entire Drop platform (FastAPI, React Native, 27-screen Operations Console, and 704 automated test invariants).



## Field Operations & Merchant Logistics

### Head of Vendor Onboarding & Rider Fleets

Deep operational execution across Nairobi peri-urban retail corridors (Ngong, Rongai, Dagoretti). Direct relationships with local water borehole operators, refill shop owners, and motorbike rider sacco cooperatives.



## Growth & Regulatory Strategy

### Head of Commercial Growth & Compliance

Specialist in East African mobile money ecosystems, Water Services Regulatory Board (WASREB) standards, and Kenya Data Protection Act compliance. Leading the commercial pipeline for schools, offices, and bulk water buyers.

📍 Local Moat: Direct grassroots ground presence and pre-agreed pilot partnerships with 30+ water refill kiosks and local motorbike stages in the Ngong-Rongai municipal corridor.



# The Ask: Proof-of-Market Seed Round

## Raising KSH 15M (~\$115,000 US Dollars / USD)

18-month operational runway to reach 25,000 monthly orders and prove recurring profitability across 5 Nairobi corridors.

40%

### Field Operations & Growth

Vendor onboarding, rider recruitment, consumer activations, and neighborhood corridor density in Nairobi satellites.

25%

### Engineering & Experience

Google Play Store listing, iOS development, real-time push dispatch, automated SMS fallbacks, and offline resilience.

20%

### Infrastructure & Float

High-availability PostgreSQL database, Redis clusters, AWS S3 storage, Clerk identity, and automated M-Pesa escrow float.

15%

### Legal & Compliance

Regulatory compliance, Water Services Regulatory Board (WASREB) standards, Kenya Data Protection Act, and vendor contracts.

#### 🎯 Key 18-Month Milestones:

- M3: 50 Stations & 100 Riders active in Ngong/Rongai pilot corridor
- M6: 1,000 Household customers, unit economics proven at scale
- M12: 10,000 monthly orders across 3 Nairobi sub-counties
- M18: 25,000 monthly orders, KSh 33M Annual Recurring Revenue (ARR) run-rate, cashflow sustainable



# Drop

Clean Water, Delivered to Your Door

## Let's talk.

🌐 Website: [drop-water-delivery-website.vercel.app](https://drop-water-delivery-website.vercel.app)

⚡ Admin Console: [drop-admin-five.vercel.app](https://drop-admin-five.vercel.app)

📦 GitHub: [github.com/WinterJackson/drop-water-delivery-app](https://github.com/WinterJackson/drop-water-delivery-app)

📱 Beta Android App (APK) Downloads: Available on website

Thank you 🙏